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心理学元分析入门:应用, 设计与实施

刘铮

香港中文大学(深圳)

应用心理学在读博士生



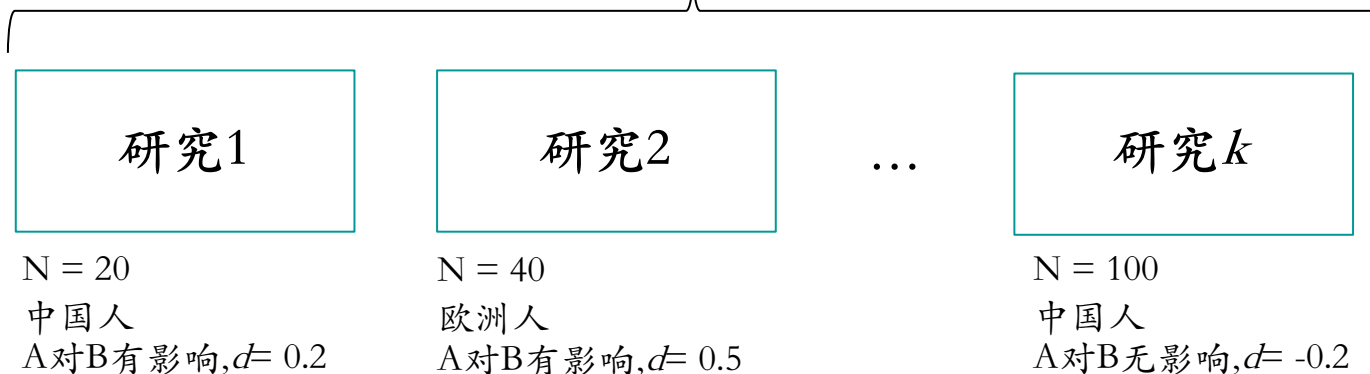
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The Chinese University of Hong Kong, Shenzhen

什么是元分析?

- 元分析是运用定量方法去概括（总结）多个研究结果的系统评价，是文献评价中，将若干个研究结果合并成一个单独数字估计的统计方法。

综合 k 个研究的证据，回答A对B到底有没有影响



传统综述/系统性综述和元分析的关系

□ 综述

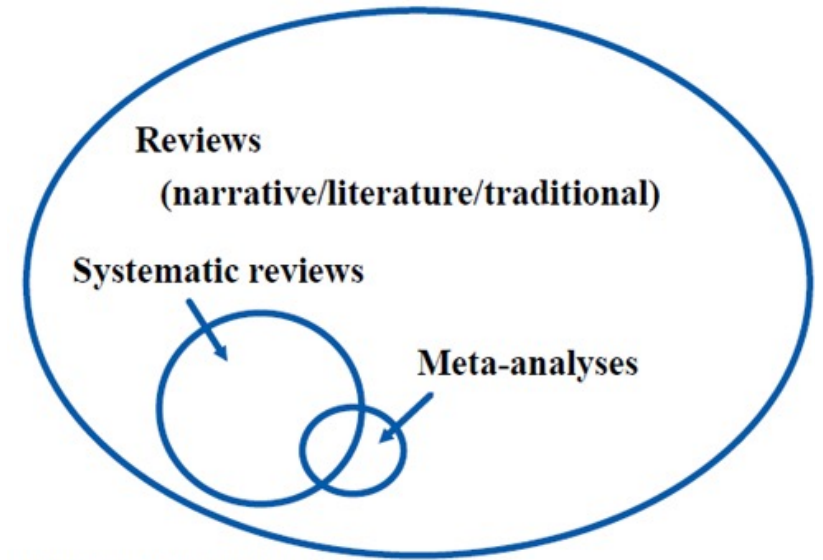
- 研究范围广
- 定性
- 无需明确定义文献选择标准

□ 系统性综述

- 通常关注单个问题
- 定性/定量
- 明确定义特定的搜索策略；列出了将包括和排除的文章

□ 元分析

- 通常关注单个问题
- 定量
- 明确定义特定的搜索策略；列出了将包括和排除的文章
- 需结合类似的研究并提取数据，计算统计学上显著性



元分析的功能

- ◆ 荟萃领域整体效应： A 对 B 是否真实有影响？
- ◆ 检验效应的边界条件： A 对 B 的影响是否受到 $CDEF..$ 的调节？

- ◆ 其他高阶功能

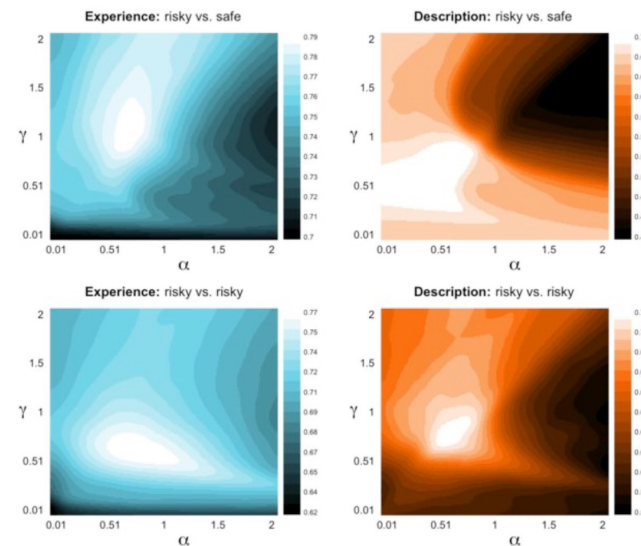
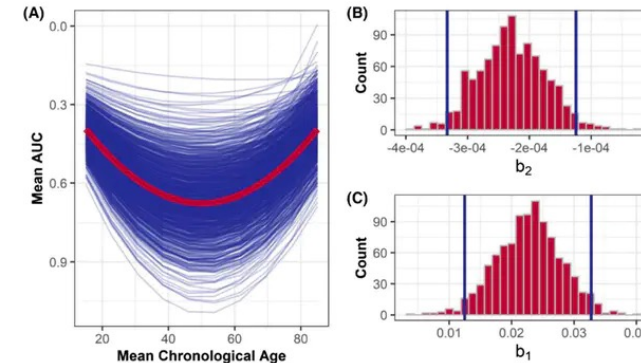


Figure 5
U-Shaped AUC Function and the Bootstrap Parameter Distributions



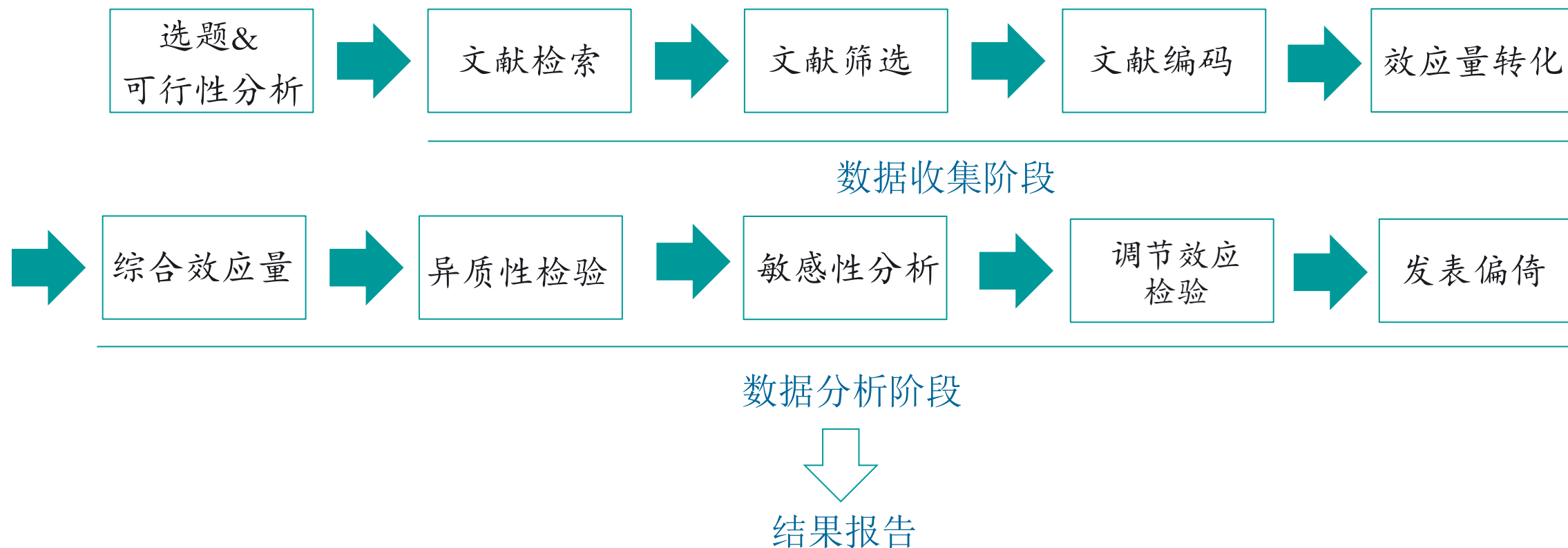
Note. The estimated AUC function. (A) U-shaped mean AUC function between mean age and delay discounting ($AUC = b_0 + b_1age + b_2age^2$). The red curve represents the recovered function based on the estimates of the RVE meta-regression. Blue curves are recovered functions based on 999 bootstrap samples. The y-axis was inverted to make the figure easier to understand in the metric of delay discount rate. (B) Distribution of b_2 with 95% CI [-0.0003, -0.0001]. The confidence intervals are marked by vertical blue lines. (C) Distribution of b_1 with 95% CI [0.012, 0.033]. AUC = area under the curve; RVE = robust variance estimation; CI = confidence interval. See the online article for the color version of this figure.

通过汇聚领域数据检验对抗的理论
(Wulff et al., 2018; Lu et.al, 2023, *Psychological Bulletin*)



元分析的基本流程

◆ “十步” 固定流程



元分析的實施



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选题

◆ 关于选题的必要思考：

➤ 研究问题是什么？

- 研究问题需要具有争议性：e.g., A对B的关系不明/不稳定。
- 研究问题需要具有探讨的价值

➤ 研究问题是否清晰？

- A与B在概念上需要存在较明确的定义
- 单篇文章不建议关注过于多元的研究设计
 - 除非，探究的重点是研究设计的调节作用



可行性分析

◆ 是否有类似题目已发表的元分析？

➤ 是。

- 在该元分析的基础上，能有多少文献的更新？
- 该元分析是否缺乏对某个关键变量的关注？
- 该元分析是否存在方法学缺陷？

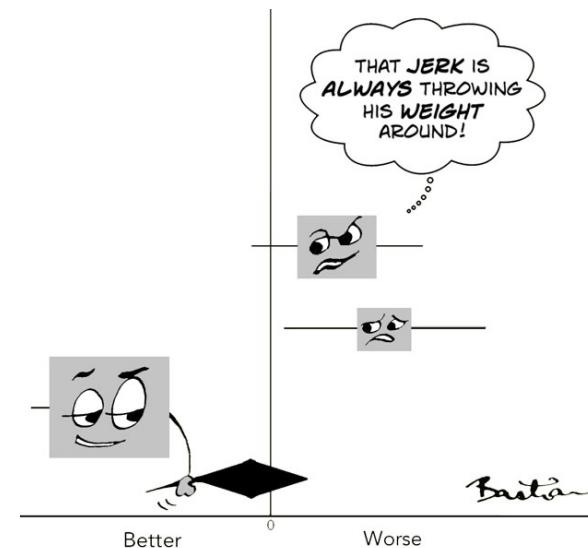
➤ 否。

◆ 该题目下大致存在多少潜在文献？

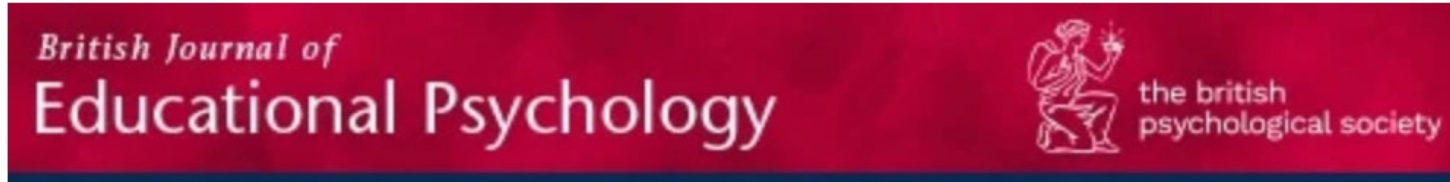
➤ 潜在的文献数量会极大程度影响元分析可靠性和发表档次！

- <10 不建议做（效应量极端值干扰很大）
- >50 是否具有足够的人力资源来完成编码工作？

✓ 粗略检索的Tip：在Google scholar检索AB关键词组合，即可快速查看是否有类似的元分析，估计相关文章的大致数量，以及，大多数研究的设计（e.g., 实验/问卷，被试内/被试间？）



实践案例



ARTICLE

The effectiveness of self: A meta-analysis of using self-referential encoding techniques in education

Zheng Liu  , Jiahui Wen , Yikang Liu , Chuan-Peng Hu 



香港中文大學(深圳)

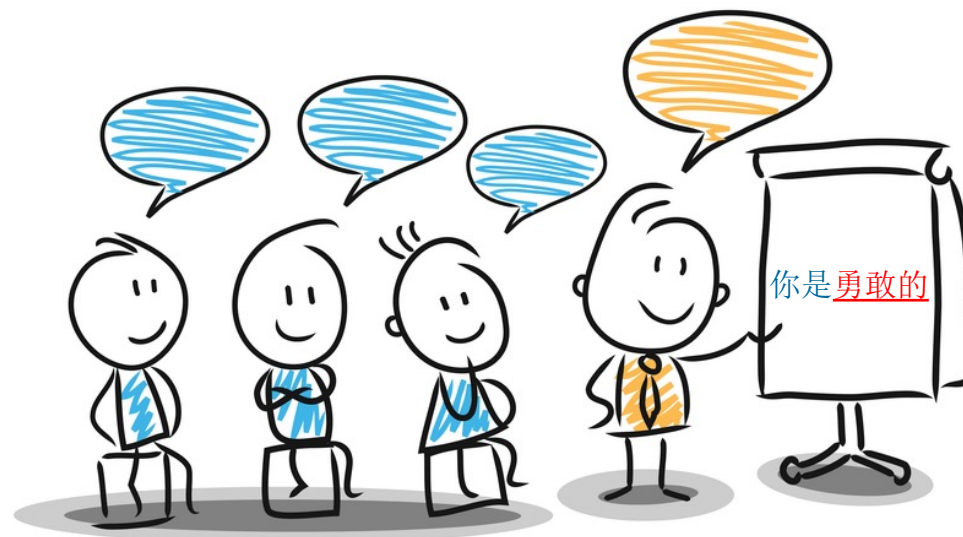
The Chinese University of Hong Kong, Shenzhen

研究背景

- ◆ 自我参照效应 (self-reference effect) 指大脑会优先加工与自我相关的信息，且这些与自我相关联的信息难以被忽视和遗忘。



鸡尾酒会效应 (Cherry, 1953)



- ◆ 自我参照编码 (self-referential encoding) 在教育教学中的应用涉及在学习材料的处理和编码过程中整合自我参照线索（如整合入自己的名字，变换人称等）。



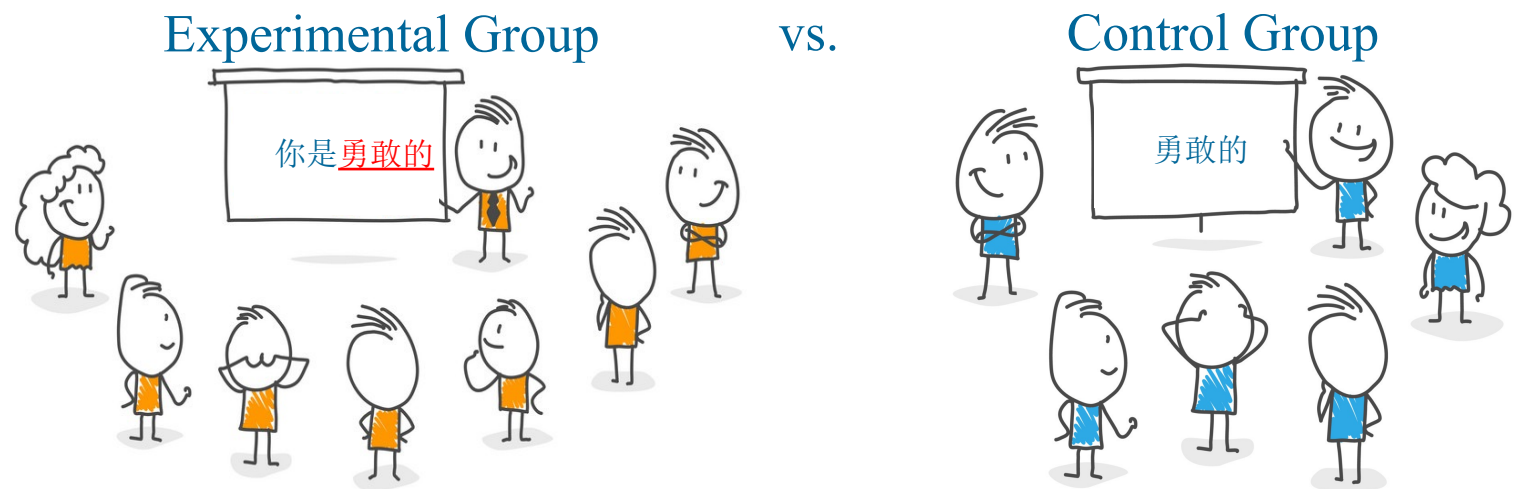
步骤一：问题提出及可行性分析

◆ 研究问题：

- 自我参照编码在教育教学中的应用的是否具有有效性？
- 这项技术在教育教学中的应用是否存在有效实施的边界？

◆ 研究设计

- 多为实验组-对照组设计，测量学习表现(e.g., 学习成绩，记忆等)



◆ 可行性分析

- 认知和教育的新交叉领域，无相似主题元分析，文献数量>10



步骤二：文献检索-确定关键词及数据库

◆ 关键词选取（尽可能多的纳入近义词）

- 自我参照编码相关：“self-referential” OR “self-reference” OR “self-bias” OR “self-tagging” OR “self-reference effect” OR “self-prioritisation effect” OR “self-prioritization effect” OR “ownership bias” OR “self-processing bias”
- 学习相关：“learning” OR “information encoding” OR “memory” OR “information retrieval”
- 教育教学相关：“education” OR “curriculum” OR “pedagogy”

◆ 数据库选取 (建议>3个，尽量纳入领域认可的数据库)

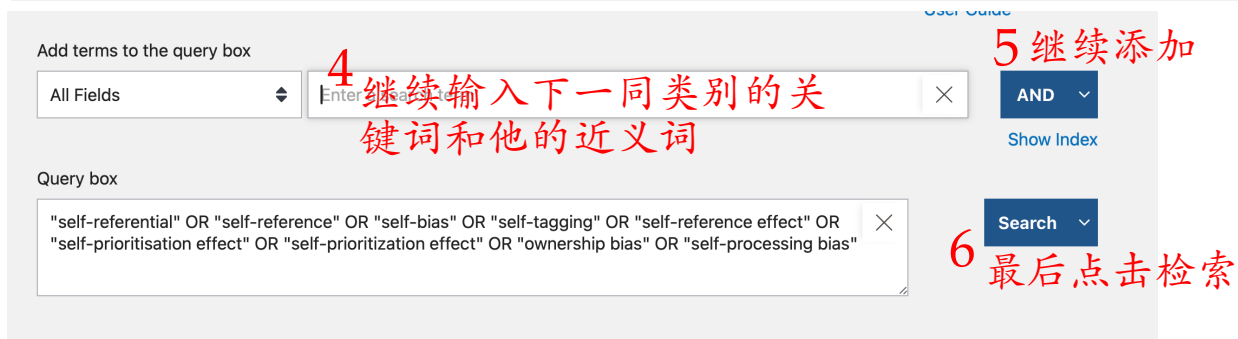
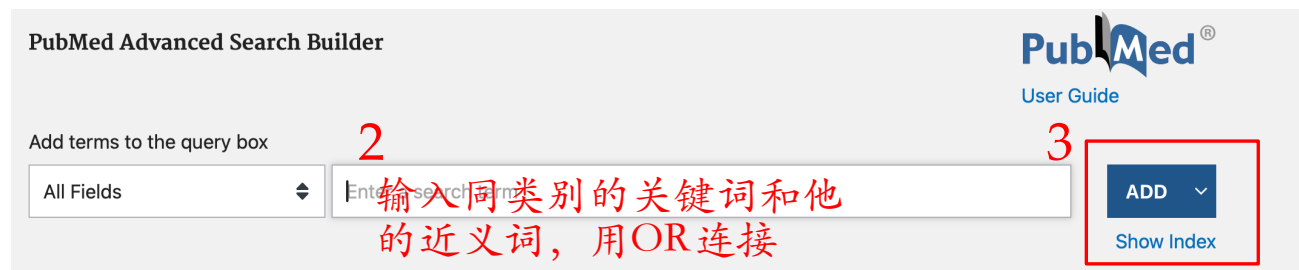
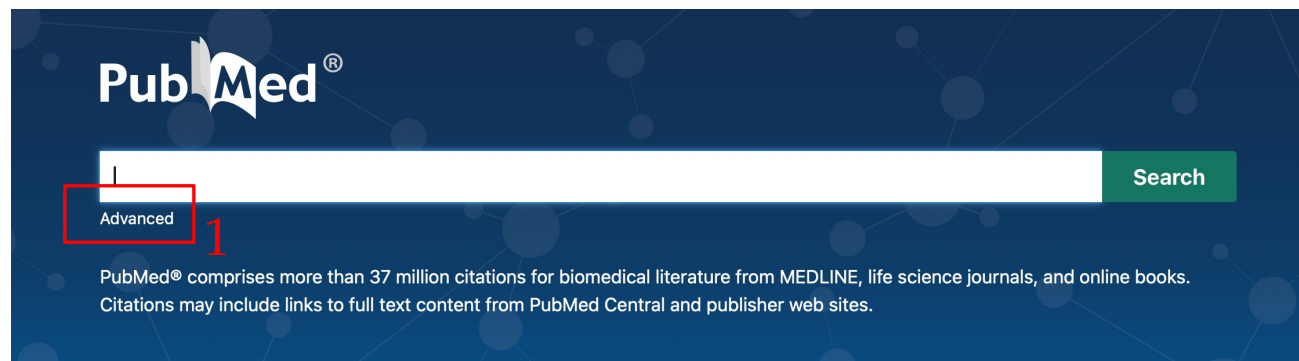
- Web of Science, PubMed, PsycINFO via Ovid, Scopus, and Science Direct

◆ 补充检索

- 一般至少需包含一种：Backward Search, Forward Search
- Backward Search: Ginns, P., Martin, A.J. & Marsh, H.W. Designing Instructional Text in a Conversational Style: A Meta-analysis. Educ Psychol Rev 25, 445–472 (2013).



数据库检索-PubMed为例



数据库检索-PubMed为例

Save Email **Send to** 7 Sort by: Best match Display options

MY NCBI FILTERS

RESULTS BY YEAR

1988 2024

TEXT AVAILABILITY

☐ Abstract

51 results

Clipboard

My Bibliography

Collections

Citation manager 8

and in phrase index: "self-processing bias"

of **Self-Referential Stimuli in Working Memory.**

Egner T.

Psychol Sci. 2019 Mar;30(3):415-423. doi: 10.1177/0956797618818483. Epub 2019 Jan 17.

PMID: 30653399

Share People preferentially attend to external stimuli that are related to themselves compared with others. Whether a similar **self-reference bias** applies to internal representations, such as those maintained in working **memory** (WM), is presently unknown. ...T ...

SPE_Edu_meta.enl

Clarivate Analytics

My Library 9

All References 1546

Configure Sync...

Recently Added

Author	Year	Title
Acevedo, Bianca P.; Aron, Arth...	2012	Neural correlates...
Adams, Shelah A.; Riggs, She...	2008	An Exploratory S...
Addis, Donna Rose a; Sacche...	2009	Episodic simulat...
Addis, Donna Rose a b; Wong...	2007	Remembering th...

Rating

Author

Acevedo, Bianca P.
Aron, Arthur



步骤三：文献筛选-文献纳入标准

◆ 对文献的要求：

- The research must be published in **English**.

◆ 对主题的要求：

- The research must examine the use of self-referential encoding techniques **in an educational context**, such as literature learning, new vocabulary learning, and problem-solving.
- The participants in the research need to be **healthy** human subjects. We do not include clinical groups in the current study.

◆ 对研究设计的要求：

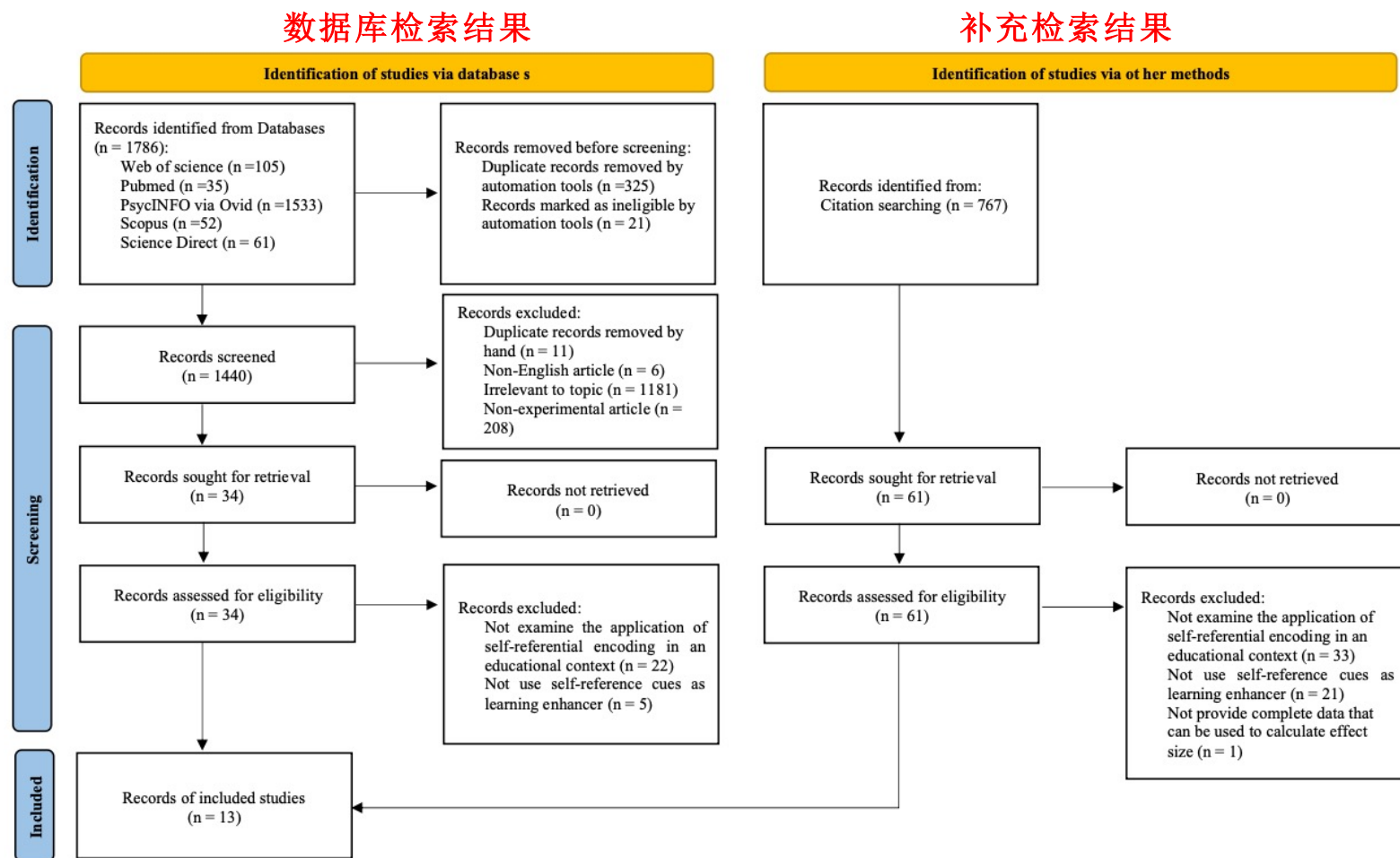
- The research must be empirical research that focuses on the application of self-referential encoding techniques **under independent comparison conditions**.

◆ 对测量的要求

- The study must include data on students' **learning process/performance**. In this meta-analysis, we define the learning process as the cognitive and psychological activities individuals engage in when acquiring new knowledge or skills.
- The study must report the measurement tools and research methods, and provide **complete data** (sample size, mean, standard deviation, t-test results, F-test results, etc.) that can be used to calculate the effect size.



PRISMA 流程图展示



<https://www.prisma-statement.org/prisma-2020-flow-diagram>



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步骤四:文献编码

◆ 编码本(Codebook)的设计

- 关于研究的基本信息:
 - 作者姓名, 年份, 期刊名, 发表状态
- 被试特征信息:
 - 被试数量, 性别比等
- 可能存在的调节变量的信息:
 - 和理论相关, 一般会考虑人口学变量, 实验设计, 测量手段等
- 统计结果
 - 与效应量相关的统计结果

◆ 编码一致性

- 需要至少两人进行背靠背编码, 并计算一致性系数
 - Cronbach's alpha
 - 结果相似性 (%)



数据摘录结果展示

样本信息

TABLE 1 Characteristics of studies included in the meta-analysis.

调节变量信息

文章信息		Proportion of female	Mean age	结果变量信息				
Author and publication year	N			Indicators of learning results	Cohort of students	Valence of materials	Form of learning materials	Research context
Axelsson et al. (2018)	23	43.48%	3.08	Accuracy	1	1	1	1
Cunningham et al. (2018), S1	30	53.33%	7.11	Memory of shape	1	1	—	1
Cunningham et al. (2018), S2	84	47.62%	7.96		1	1	—	1
d'Ailly et al. (1997)	100	50.00%	—	Reaction time	1	1	1	1
Ditman et al. (2010), S1	36	50.00%	19.11	Reaction time, d-prime	3	1	1	1
Dutke et al. (2016)	59	59.32%	16	Test scores of comprehensions and transfer	2	1	2	2
Ginns and Fraser (2010)	40	77.50%	22.60	Test scores	3	1	2	1
Halpin et al. (1984)	49	59.18%	8.21	Orienting task results	1	1	2	1
Kühl and Zander (2017), S1	77	77.92%	23.42	Test scores of knowledge test (factual and transfer)	3	2	1	1
Kühl and Zander (2017), S2	71	78.87%	21.56		3	2	1	1
Kühl and Münzer (2021), G1	65	—	22.30	Test scores of knowledge test (retention and transfer)	3	2	1	1
Kühl and Münzer (2021), G2	60	—			3	2	1	1
Li et al. (2021), S1	48	100.00%	18.81	Reaction time	3	1	1	1
Li et al. (2021), S2	29	100.00%	18.69		3	1	1	1
Mayer et al. (2004), S1	62	64.52%	18.70	Scores on retention and transfer tests	3	1	2	1
Mayer et al. (2004), S2	27	40.74%	18.60		3	1	2	1
Mayer et al. (2004), S3	32	21.88%	20.20		3	1	2	1
Sinatra et al. (2016)	111	53.15%	19.94	Time spent solving the standard puzzle	3	1	1	1
Turk et al. (2015), S1	47	57.45%	8.10	Sentence length, spelling accuracy	1	1	2	2
Turk et al. (2015), S2	32	53.13%	8.43		1	1	2	2

Note: S with numbers represents different studies from a single article; G with numbers represents independent groups from a single study; N represents sample size; in cohort of students section, 1—primary and below, 2—secondary and high school, 3—university level and above; in research context section, 1—laboratory, 2—classroom; in form of learning materials section, 1—computer-based, 2—paper-based; in valence of materials section, 1—negative, 2—non-negative.



元分析数据处理所需R包

- 数据清洗: tidyverse
- 单个研究效应量计算: esc
- 效应量合并等分析: metafor, meta

```
# Packages
if (!requireNamespace('pacman', quietly = TRUE)) {
  install.packages('pacman')
}

pacman::p_load(tidyverse, esc, metafor, meta)
```



步骤五:单个研究效应量转化

◆ 什么是效应量?

- 效应量是衡量实验效应强度或者变量关联强度的指标, 它不受样本容量大小的影响(或者影响很小)。

◆ 效应量的种类 (郑昊敏 等, 2011)

➤ 差异类(difference-type)

- 一般用于实验研究, 衡量两组均值比较或多组均值的差异程度。
- e.g., Cohen's d , Hedge's g

➤ 相关类效应量 (correlation-type)

- 相关类效应量一般用于变量相关的研究中, 其大小衡量了两个或多个变量共变的程度。
- e.g., r , β

➤ 组重叠效应量 (group-overlap) 心理学研究很少出现

- 在均值比较中, 假定单变量的两组被试得分分布相似(都呈正态分布), 当两组分布重叠程度很小时, 可以说这两组均值之间存在显著差异。



效应量的计算-独立样本

Reported	Computation of Needed Quantities
$\bar{Y}_1, \bar{Y}_2, S_{Pooled}, n_1, n_2$	$d = \frac{Y_1 - Y_2}{S_{Pooled}}, \quad v = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$
t, n_1, n_2	$d = t \sqrt{\frac{n_1 + n_2}{n_1 n_2}}, \quad v = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$
F, n_1, n_2	$d = \pm \sqrt{\frac{F(n_1 + n_2)}{n_1 n_2}}, \quad v = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$
$p(\text{one-tailed}), n_1, n_2$	$d = \pm t^{-1}(p) \sqrt{\frac{n_1 + n_2}{n_1 n_2}}, \quad v = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$
$p(\text{two-tailed}), n_1, n_2$	$d = \pm t^{-1}\left(\frac{p}{2}\right) \sqrt{\frac{n_1 + n_2}{n_1 n_2}}, \quad v = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$

Cooper et al. (2009)



效应量的计算-配对样本

Reported	Computation of Needed Quantities
$\bar{Y}_1, \bar{Y}_2, S_{Difference}, r, n$ (number of pairs)	$d = \left(\frac{\bar{Y}_1 - \bar{Y}_2}{S_{Difference}} \right) \sqrt{2(1-r)}, \quad v = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r)$
t (from paired t -test), r, n	$d = t \sqrt{\frac{2(1-r)}{n}}, \quad v = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r)$
F (from repeated measures ANOVA), r, n	$d = \pm \sqrt{\frac{2F(1-r)}{n}}, \quad v = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r)$
p (one-tailed), r, n	$d = \pm t^{-1}(p) \sqrt{\frac{2(1-r)}{n}}, \quad v = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r)$
p (two-tailed), r, n	$d = \pm t^{-1}\left(\frac{p}{2}\right) \sqrt{\frac{2(1-r)}{n}}, \quad v = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r)$

Cooper et al. (2009)



效应量的计算-相关类研究

Reported	Computation of Needed Quantities
r, n	$v_r = \frac{(1-r^2)^2}{n-1}, \quad z = 0.5 \ln\left(\frac{1+r}{1-r}\right), \quad v_z = \frac{1}{n-3}$
t, n	$r = \pm \sqrt{\frac{t^2}{t^2 + n - 2}}, \quad v_r = \frac{(1-r^2)^2}{n-1}, \quad z = 0.5 \ln\left(\frac{1+r}{1-r}\right), \quad v_z = \frac{1}{n-3}$
t, r	$n = t^2\left(\frac{1-r^2}{r^2}\right) - 2, \quad v_r = \frac{(1-r^2)^2}{n-1}, \quad z = 0.5 \ln\left(\frac{1+r}{1-r}\right), \quad v_z = \frac{1}{n-3}$
$p(\text{one-tailed}), r$	$n = [t^{-1}(p)]^2\left(\frac{1-r^2}{r^2}\right) - 2, \quad v_r = \frac{(1-r^2)^2}{n-1}, \quad z = 0.5 \ln\left(\frac{1+r}{1-r}\right), \quad v_z = \frac{1}{n-3}$
$p(\text{two-tailed}), r$	$n = \left[t^{-1}\left(\frac{p}{2}\right)\right]^2\left(\frac{1-r^2}{r^2}\right) - 2, \quad v_r = \frac{(1-r^2)^2}{n-1}, \quad z = 0.5 \ln\left(\frac{1+r}{1-r}\right), \quad v_z = \frac{1}{n-3}$

Cooper et al. (2009)



单个研究效应量转化

◆ 报告Mean和SD的效应量的计算

Dutke et al. (2016): ... The final sample comprised n=29 students in the personalized text condition and n=30 students in the standard text condition.

Table 1 Comprehension and transfer performance: means and standard deviations

		Type of text			
		Standard		Personalized	
		Study	Restudy	Study	Restudy
Comprehension	M	87.1	88.4	87.0	89.0
	SD	7.6	8.1	7.3	7.5

ES for comprehension-study:

```
esc_mean_sd(  
  grp1m = 87, grp1sd = 7.3, grp1n = 29,  
  grp2m = 87.1, grp2sd = 7.6, grp2n = 30,  
  es.type="g")
```

◆ 报告Mean和SE的效应量的计算

```
esc_mean_se(  
  grp1m = 87, grp1se = 7.3, grp1n = 29,  
  grp2m = 87.1, grp2se = 7.6, grp2n = 30,  
  es.type="g")
```

Effect Size Calculation for Meta Analysis

Conversion: mean and sd to effect size Hedges' g

Effect Size: -0.0132

Standard Error: 0.2604

Variance: 0.0678

Lower CI: -0.5236

Upper CI: 0.4972

Weight: 14.7454



单个研究效应量转化

◆ 报告t检验的效应量的计算

Ginns and Fraser (2010) : “The personalized group scored higher on the terminology test than the non-personalized group, $t(38) = 2.98$, $p = 0.005$, $d = 0.92$ [0.27–1.58].”

```
esc::esc_t(t = 2.98, #t值  
  totaln = 40, #总被试数  
  es.type="g") #效应量的种类
```

Effect Size Calculation for Meta Analysis

Conversion: t-value to effect size Hedges' g

Effect Size:	0.9236
Standard Error:	0.3333
Variance:	0.1111
Lower CI:	0.2703
Upper CI:	1.5769
Weight:	9.0009



单个研究效应量转化

◆ 报告非标准化回归系数的效应量的计算

```
esc::esc_B(  
  b=3.3, #非标准化回归系数  
  sdy=5, #因变量的SD  
  grp1n = 100, #第一组的人数  
  grp2n = 150, #第二组的人数  
  es.type = "g") #效应量的种类
```

Effect Size Calculation for Meta Analysis

Conversion: unstandardized regression coefficient to effect size Hedges' g

Effect Size:	0.6941
Standard Error:	0.1328
Variance:	0.0176
Lower CI:	0.4338
Upper CI:	0.9544
Weight:	56.7018

◆ 报告标准化回归系数的效应量的计算

```
esc_beta(  
  beta=0.7, #标准化回归系数  
  sdy=3, #因变量的SD  
  grp1n=100, #第一组的人数  
  grp2n=150, #第二组的人数  
  es.type = "g") #效应量的种类
```

Effect Size Calculation for Meta Analysis

Conversion: standardized regression coefficient to effect size Hedges' g

Effect Size:	1.9868
Standard Error:	0.1569
Variance:	0.0246
Lower CI:	1.6793
Upper CI:	2.2942
Weight:	40.6353



步骤六:综合效应量(Overall Effect Size)

- ◆ 对效应量进行合并是为了获得研究的整体效应大小。
- ◆ 常用的两种模型
 - 固定效应模型(Fixed-Effect Model)
 - 假设所有的研究及其效应大小都来自单一的同质群体(Borenstein et al., 2011)。
 - 随机效应模型(Random-Effects Model)
 - 异质群体选用随机效应模型。
- ◆ 元分析综合效应量的解读(Cohen, 1988, 现存在较大争议)
 - ~ 0.2 小效应
 - ~ 0.5 中等效应
 - ~ 0.8 大效应



加载需要的效应量表并预处理

```
#import
effect_size <- read.csv("../datasheets/effect_size.csv") %>%
  dplyr::select(
    id, #独立研究编号
    Information, #作者信息
    N, #被试数
    ES, #刚刚计算的每个研究的效应量
    VI, #刚刚计算的, 效应量的variance
    "Cohort" = subgroup1,
    "Context" = subgroup2, #纳入的调节变量
    "Materials" = subgroup3,
    "Valence" = subgroup4
  ) %>%
  dplyr::filter(row_number() %in% c(1:20))
head(effect_size)
```

	id	Information	N	ES	VI	Cohort	Context	Materials	Valence
	<int>	<chr>	<int>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<chr>
1	1	Axelsson et al. (2018)	23	0.4351	0.0489	1	1	1	1
2	2	Cunningham et al. (2018), S1	30	1.3587	0.0827	1	1		1
3	3	Cunningham et al. (2018), S2	84	0.8247	0.0491	1	1		1
4	4	d'Ailly et al. (1997)	100	0.2108	0.0096	1	1	1	1
5	5	Ditman et al. (2010), S1	36	0.4210	0.0564	3	1	1	1
6	6	Dutke et al. (2016)	59	0.5690	0.0732	2	2	2	1



对效应量表格进行格式转换

#数据预处理

```
df <- metafor::escalc( #转换成escalc的格式
  measure="SMD",
  data = effect_size,
  yi= ES, #指定每个研究的效应量是哪列
  vi = VI, #指定每个研究的效应量的variance是哪列
  slab = paste("Study ID:", id)) #注明研究的label
```

#按效应量大小进行排序，方便后续画图展示

```
df <- df[order(df$yi), ]
head(df)
```

得到从小到大排列的效应量

id	Information	N	ES	VI	Cohort	Context	Materials	Valence	yi	vi
12	Kühl and Münzer (2021), G2	60	-0.5278	0.0691	3	1	1	2	-0.5278	0.069
10	Kühl and Zander (2017), S2	71	-0.4687	0.0583	3	1	1	2	-0.4687	0.058
9	Kühl and Zander (2017), S1	77	-0.0640	0.0530	3	1	1	2	-0.0640	0.053
18	Sinatra et al. (2016)	111	-0.0560	0.0541	3	1	1	1	-0.0560	0.054
14	Li et al. (2021), S2	29	0.0477	0.0276	3	1	1	1	0.0477	0.027
4	d'Ailly et al. (1997)	100	0.2108	0.0096	1	1	1	1	0.2108	0.009
15	Mayer et al. (2004), S1	62	0.2529	0.0658	3	1	2	1	0.2529	0.065
11	Kühl and Münzer (2021), G1	65	0.3626	0.0629	3	1	1	2	0.3626	0.062
16	Mayer et al. (2004), S2	27	0.3687	0.1580	3	1	2	1	0.3687	0.158
5	Ditman et al. (2010), S1	36	0.4210	0.0564	3	1	1	1	0.4210	0.056
20	Turk et al. (2015), S2	32	0.4292	0.0640	1	2	2	1	0.4292	0.064
1	Axelsson et al. (2018)	23	0.4351	0.0489	1	1	1	1	0.4351	0.048
8	Halpin et al. (1984)	49	0.5089	0.0185	1	1	2	1	0.5089	0.018
6	Dutke et al. (2016)	59	0.5690	0.0732	2	2	2	1	0.5690	0.073
17	Mayer et al. (2004), S3	32	0.5783	0.1316	3	1	2	1	0.5783	0.131
19	Turk et al. (2015), S1	47	0.7039	0.0907	1	2	2	1	0.7039	0.090
3	Cunningham et al. (2018), S2	84	0.8247	0.0491	1	1		1	0.8247	0.049
7	Ginns and Fraser (2010)	40	0.9236	0.1111	3	1	2	1	0.9236	0.111
2	Cunningham et al. (2018), S1	30	1.3587	0.0827	1	1		1	1.3587	0.082
13	Li et al. (2021), S1	48	1.4537	0.0529	3	1	1	1	1.4537	0.052



步骤六:综合效应量(Overall Effect Size)

◆ 使用Metafor::rma

```
res <- metafor::rma(  
  yi, #每个研究的效应量  
  vi, #每个研究的效应量的variance  
  method = "REML", #method="REML" is the default  
  data = df)  
res
```

Random-Effects Model (k = 20; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.1917 (SE = 0.0820)
tau (square root of estimated tau^2 value): 0.4379
I^2 (total heterogeneity / total variability): 80.32%
H^2 (total variability / sampling variability): 5.08

Test for Heterogeneity:
Q(df = 19) = 81.4191, p-val < .0001

Model Results:

Hedge's g	SE	p value		95% CI	
estimate	se	zval	pval	ci.lb	ci.ub
0.4036	0.1128	3.5791	0.0003	0.1826	0.6247 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

对综合效应量的报道:

The overall effect size **Hedge's g is .40, SE=.11, $p<.001$, 95% CI [.18, .62]**, indicating that applying self-referential encoding techniques is effective and has a **small to moderate effect** on learning.



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步骤六:综合效应量(Overall Effect Size)

◆ 使用Meta::metagen

```
res1 <- meta::metagen(  
  TE = yi, #每个研究的效应量  
  seTE = sqrt(vi), #每个研究的效应量的标准误  
  data = df,  
  studlab = df$id,  
  fixed = FALSE,  
  random = TRUE, #选择随机效应模型  
  prediction = TRUE, #是否需要CI  
  sm = "SMD")  
res1
```

The overall effect size Hedge's g is .40,
 $SE=.11$, $p<.001$, 95% CI [.18, .62].

Number of studies: $k = 20$

	SMD	95%-CI	z	p-value
Random effects model	0.4036	[0.1826; 0.6247]	3.58	0.0003
Prediction interval		[-0.5463; 1.3536]		

Quantifying heterogeneity:

$\tau^2 = 0.1917$ [0.0848; 0.4818]; $\tau = 0.4379$ [0.2912; 0.6941]
 $I^2 = 76.7\%$ [64.2%; 84.8%]; $H = 2.07$ [1.67; 2.56]

Test of heterogeneity:

Q	d.f.	p-value
81.42	19	< 0.0001

Details on meta-analytical method:

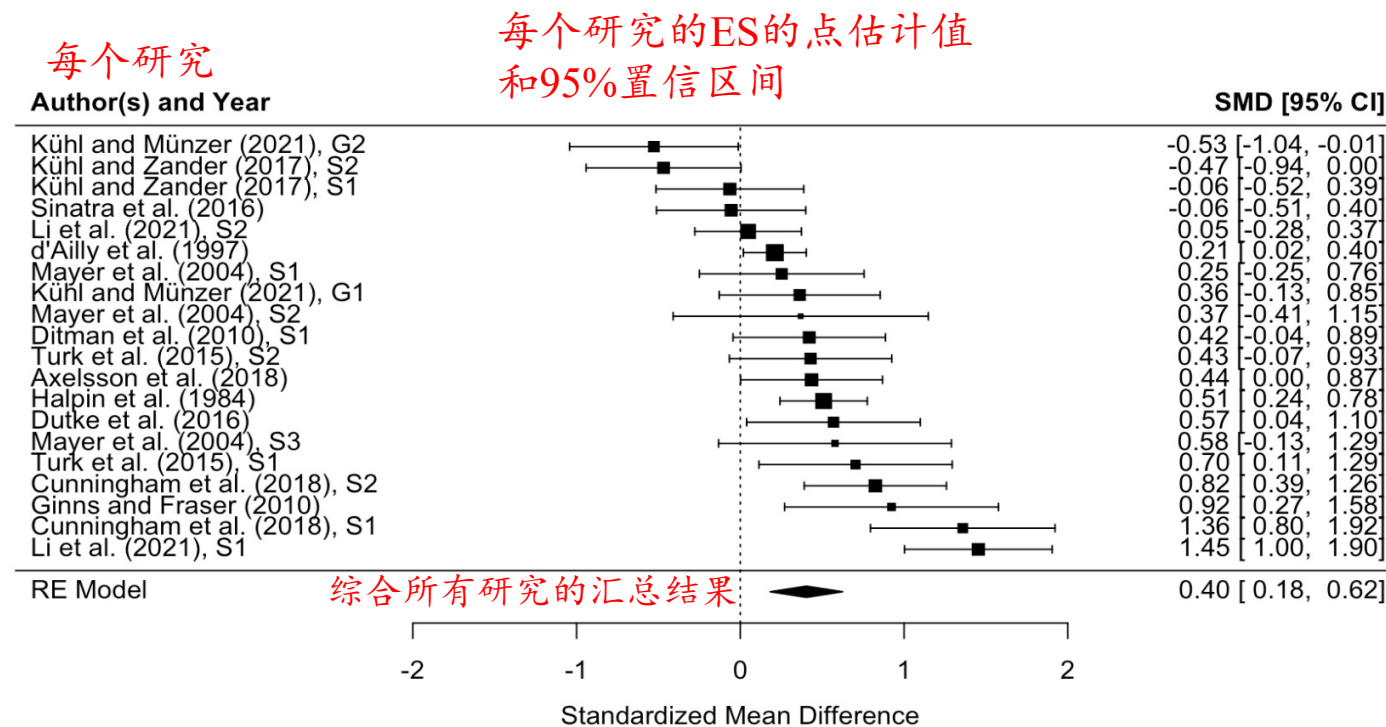
- Inverse variance method
- Restricted maximum-likelihood estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Prediction interval based on t-distribution ($df = 18$)



森林图(Forest Plot)

- 森林图是用于展示多个研究的效应量及其置信区间，以直观地展示各研究结果的一致性和差异性，以及合并的效应量的大小。

```
metafor::forest(  
  res,  
  slab = paste(df$Information),  
  header="Author(s) and Year" )
```



步骤七:异质性检验

- ◆ 描述参与者、干预措施和一系列研究间测量结果的差异和多样性，或那些研究间的内在真实性的变异。
 - 当存在异质性时，才能表明随机效应模型选择合理，且后续进行亚组分析和元回归才有意义（因为需要判定哪些可能的调节变量贡献了异质性）
- ◆ 判断异质性的Q检验和 I^2 检验：

Random-Effects Model (k = 20; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.1917 (SE = 0.0820)

tau (square root of estimated tau^2 value): 0.4379

I^2 (total heterogeneity / total variability): 80.32%

H^2 (total variability / sampling variability): 5.08

Test for Heterogeneity:

Q(df = 19) = 81.4191, p-val < .0001

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub
0.4036	0.1128	3.5791	0.0003	0.1826	0.6247 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

对异质性的报道：

There was a strong evidence of homogeneity among studies, $Q(19)=81.42, p<.001, I^2 = 80.32\%$. The I^2 value was greater than 75%, indicating high heterogeneity among studies, and it is reasonable to select the random-effect model.



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步骤八:敏感性分析 (Sensitivity Analysis)

- ◆ 用来评估元分析的结果稳健性的分析。
- ◆ 常用方法:
 - “去一法” (Leave-One-Out-method)
 - 每次排除一项研究，重新计算元分析效应量 $K-1$ 次。
 - 检测对元分析的结果影响最大的研究，以及评估这种影响是否会扭曲混合效应量 (Viechtbauer and Cheung, 2010)。
 - **GOSH Plot** (Olkin et al., 2012)
 - 拟合所有研究子集的元分析模型。相较于“去一法”，不仅要拟合 $K-1$ 个模型，还要拟合所有 $2k-1$ 个可能的研究组合。



步骤八:敏感性分析 (Sensitivity Analysis)

□ “去一法”

```
llo <- metafor::leave1out(res)
#可视化
metafor::forest(
  llo$estimate,
  sei = llo$se,
  header = "Ommited Study",
  slab = paste(df$Information),
  xlab = "Leave One Out Estimate",
  refline = coef(res))
```

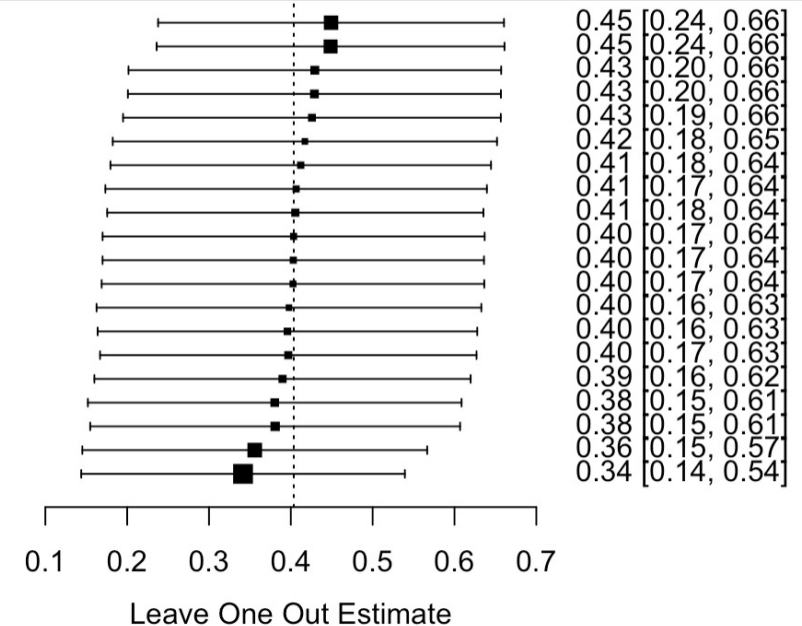
Through eliminating one study successively, the conclusions were not drastically changed in this analysis. The changes in effect size were small after the removal of any study, ranging from .34 (95% CI [.14, .54]) to .45 (95% CI [.24, .66]), indicating that the meta-analysis results were reliable.

Ommited Study

Kühl and Münzer (2021), G2
Kühl and Zander (2017), S2
Kühl and Zander (2017), S1
Sinatra et al. (2016)
Li et al. (2021), S2
d'Ailly et al. (1997)
Mayer et al. (2004), S1
Kühl and Münzer (2021), G1
Mayer et al. (2004), S2
Ditman et al. (2010), S1
Turk et al. (2015), S2
Axelsson et al. (2018)
Halpin et al. (1984)
Dutke et al. (2016)
Mayer et al. (2004), S3
Turk et al. (2015), S1
Cunningham et al. (2018), S2
Ginns and Fraser (2010)
Cunningham et al. (2018), S1
Li et al. (2021), S1

去掉该研究后的
重新估计的ES

Estimate [95% CI]



步骤九:调节效应检验-亚组分析 (Subgroup Analysis)

- ◆ 考察元分析中的不同亚组之间的差异，并确定它们在效应大小方面是否存在差异。
- ◆ E.g., 我们想考察材料附带的情感价值（负性/非负性）是否会影响教学效果：

```
valence_subgroup <- stats::update(  
  res1,  
  subgroup = Valence, #亚组标记  
  random = TRUE,  
  fixed = FALSE)  
valence_subgroup
```

The intergroup heterogeneity test showed that the effect sizes of the studies with the different valence of materials were significantly different ($Q(1)=9.65, p=.002$), suggesting the valence of materials **had a significant moderating effect** on the effectiveness of self-referential encoding.

Number of studies: k = 20

	SMD	95%-CI	z	p-value
Random effects model	0.4036	[0.1826; 0.6247]	3.58	0.0003
Prediction interval		[-0.5463; 1.3536]		

Quantifying heterogeneity:

$\tau^2 = 0.1917$ [0.0848; 0.4818]; $\tau = 0.4379$ [0.2912; 0.6941]
 $I^2 = 76.7\%$ [64.2%; 84.8%]; $H = 2.07$ [1.67; 2.56]

Test of heterogeneity:

Q	d.f.	p-value
81.42	19	< 0.0001

亚组的效应量

Results for subgroups (random effects model):

	k	SMD	95%-CI	τ^2	τ	Q	I^2
Valence = 2	4	-0.1717	[-0.5702; 0.2268]	0.1048	0.3237	8.10	63.0%
Valence = 1	16	0.5422	[0.3321; 0.7523]	0.1226	0.3501	52.31	71.3%

Test for subgroup differences (random effects model):

Q	d.f.	p-value
Between groups	9.65	1 0.0019

Details on meta-analytical method:

- Inverse variance method
- Restricted maximum-likelihood estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Prediction interval based on t-distribution (df = 18)



步骤九:调节效应检验-元回归 (Meta Regression)

- ◆ 通过寻找连续性的调节变量 (如被试量、年龄、性别比例等)来对研究结果之间的差异进行解释。
- ◆ E.g., 我们想考察被试量(N)是否影响效应量大小:

```
N_metareg <- meta::metareg(  
  res1,  
  N)
```

Mixed-Effects Model (k = 20; tau² estimator: REML)

tau ² (estimated amount of residual heterogeneity):	0.1749 (SE = 0.0788)
tau (square root of estimated tau ² value):	0.4182
I ² (residual heterogeneity / unaccounted variability):	77.27%
H ² (unaccounted variability / sampling variability):	4.40
R ² (amount of heterogeneity accounted for):	8.77%

Test for Residual Heterogeneity:
QE(df = 18) = 72.9187, p-val < .0001

Test of Moderators (coefficient 2):
QM(df = 1) = 2.6276, p-val = 0.1050



步骤十：发表偏倚 (Test of Publication Bias)

- ◆ 高效应量的研究比具有低效应量的研究更有可能被发表，并引起相应的发表偏倚。
 - 这将使得研究池中的研究的效应量均偏高，并进一步使得相应的元分析中的合并效应量可能高于其真实效应。
- ◆ 发表偏倚的检验方法：
 - 漏斗图 (Funnel plot)
 - 可视化是否缺少效应量较小的小型研究。
 - Egger Regression
 - 量化漏斗图的不对称性，并进行统计检验。

Positive results



$p < 0.05$

Negative results



$p > 0.05$



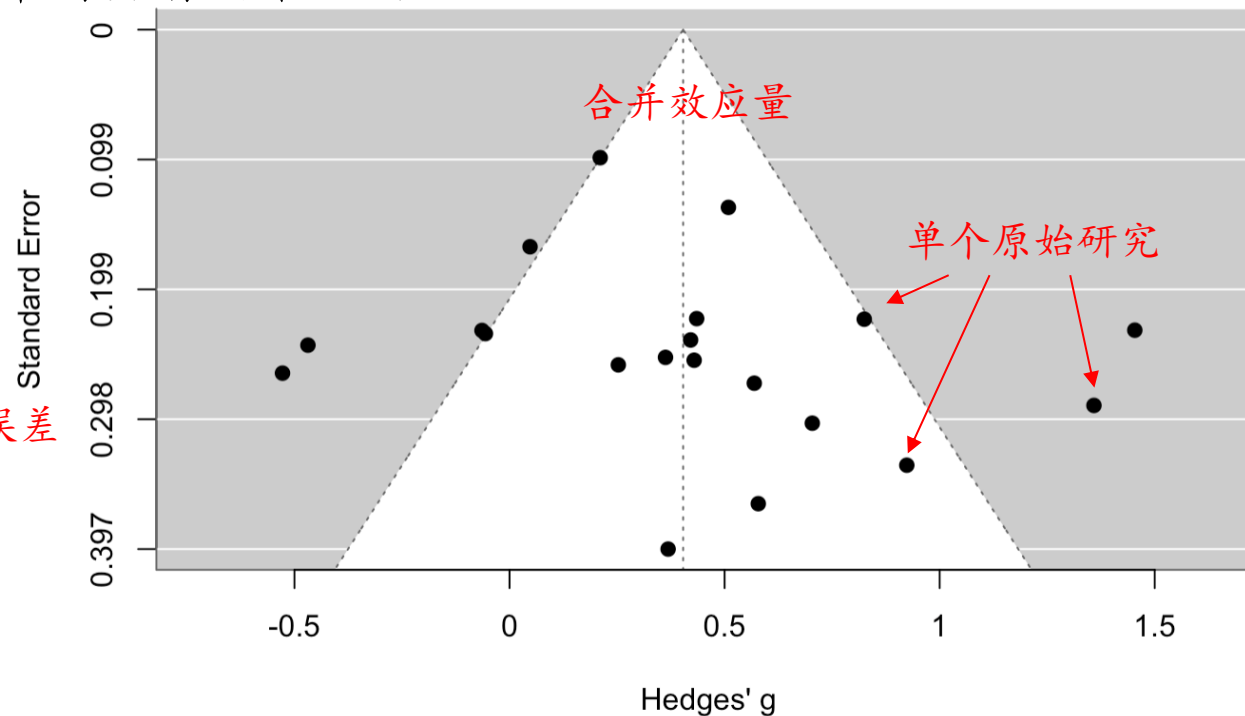
步骤十：发表偏倚 (Test of Publication Bias)

□ 漏斗图

- 图越往上方显示的是标准误较小（样本量更大）的研究。
- 漏斗图假设在没有发表偏差的情况下，所有的研究都会对称地分布在合并效应量的两侧，当存在出版偏差时，该漏斗则会看起来不对称。

```
metafor::funnel(res,  
  xlab = "Hedges' g")
```

转换后的标准误差



步骤十：发表偏倚 (Test of Publication Bias)

❑ Egger Regression

- 量化漏斗图的不对称性，并进行统计检验

```
metafor::regtest(res, model="lm")
```

Regression Test for Funnel Plot Asymmetry

Model: weighted regression with multiplicative dispersion
Predictor: standard error

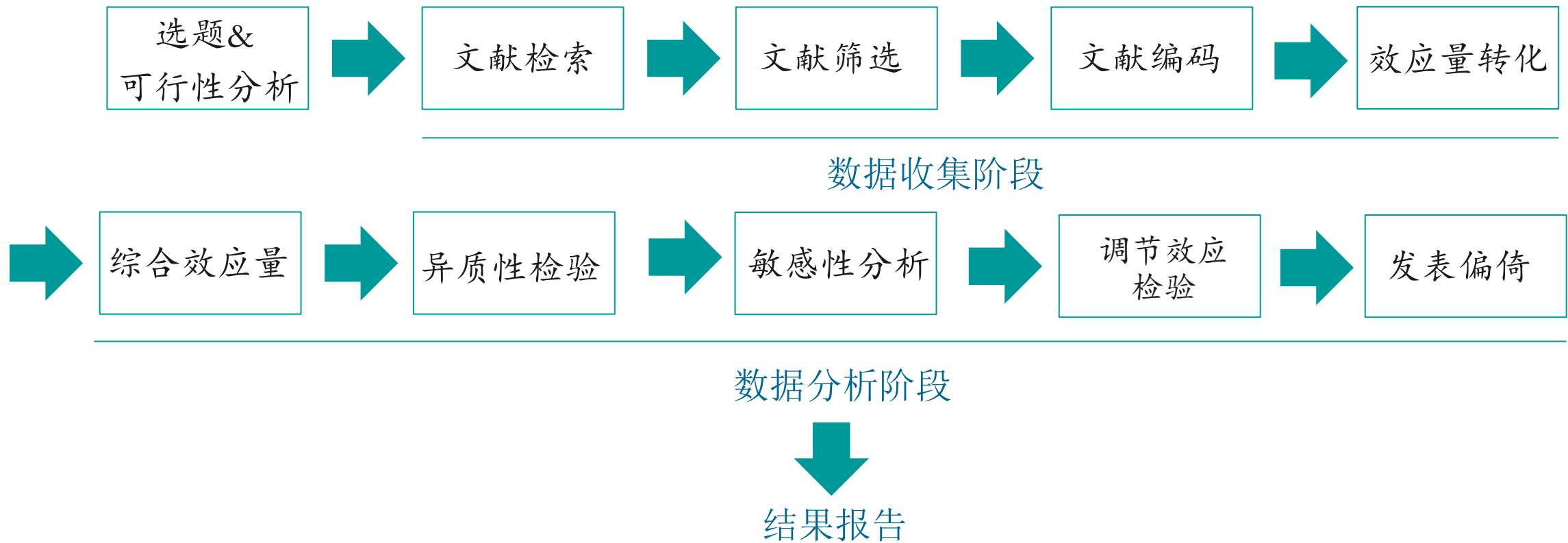
Test for Funnel Plot Asymmetry: $t = 0.9106$, $df = 18$, $p = 0.3746$
Limit Estimate (as $se_i \rightarrow 0$): $b = 0.1170$ (CI: $-0.4623, 0.6962$)

We applied Egger linear regression method to further evaluate whether publication bias exists. The result shows that the funnel diagram is symmetric $t(18)=.91, p=.375$.



回顾元分析的基本流程

◆ “十步” 固定流程



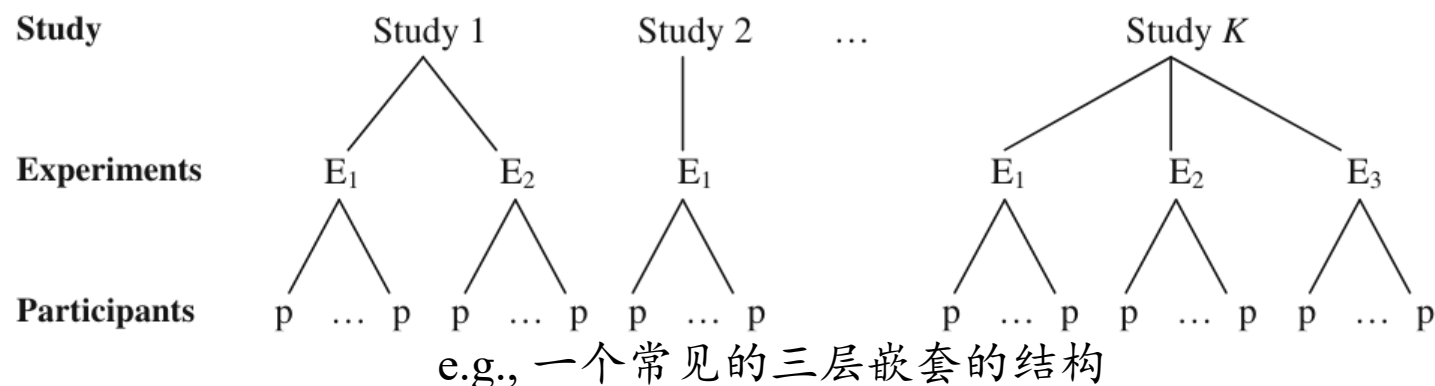
传统元分析的局限性

◆ 传统元分析的前提：效应量之间具有相互独立性



◆ 然而：

- 一项研究可能报告多个效应量
- 同一研究下的效应量可能具有内部高相关性 (Lipsey and Wilson, 2001)



◆ 传统解决方法：

- 直接忽略非独立性，假设存在的ES都是相互独立的
- 将同一个研究中的效应量进行简单平均



多水平元分析(Multi-level Meta Analysis)

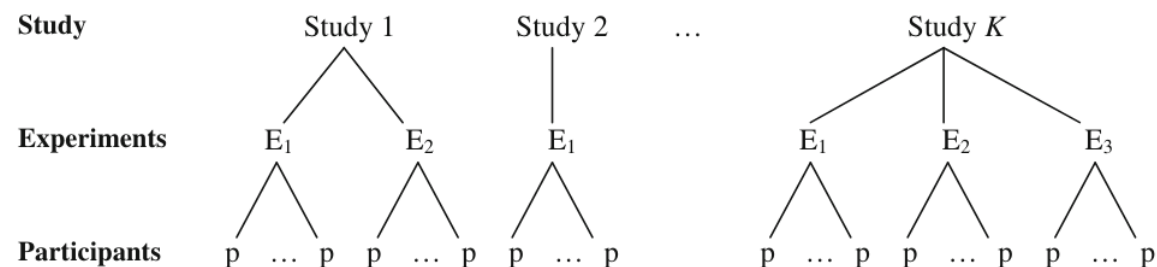
◆ 分解方差

◆ 传统元分析将方差来源分为两个个水平

- 水平1——抽样方差
- 水平2——研究间方差

◆ 三水平元分析将方差来源分为三个个水平

- 水平1——抽样方差
- 水平2——研究内方差
- 水平3——研究间方差



对元分析的反思

❑ “垃圾进，垃圾出”

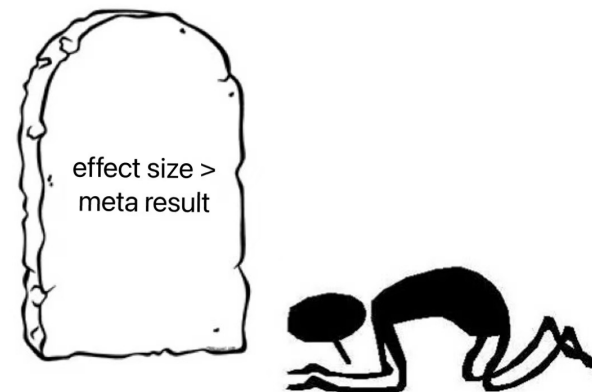
- 如果元分析中包含了许多低质量的研究，那么原始研究中的基本错误将会转移到元分析中，甚至由于元分析的影响力，这些错误将更难识别。
- **对纳入文章进行质量筛选至关重要。**

❑ 苹果和橘子问题

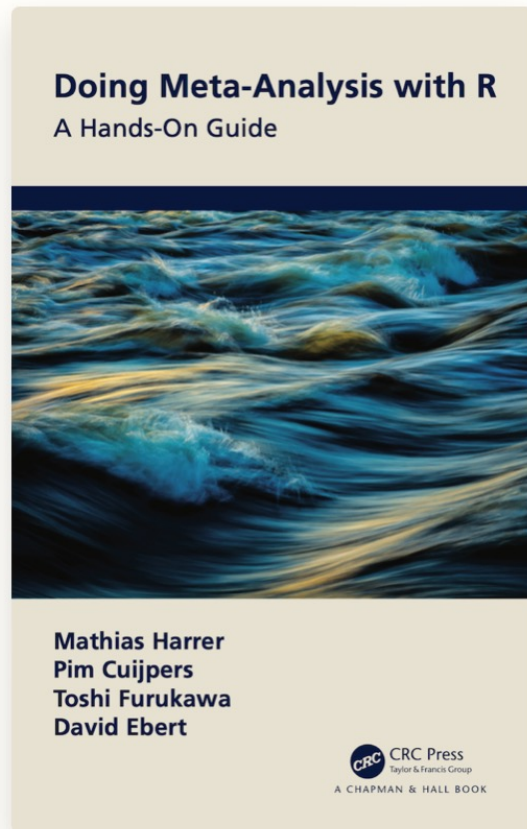
- 元分析研究的概念过于宽泛，有时会将完全不同类别的研究进行整合。
- **需要明确变量的理论定义和操作性定义。**

❑ 对效应量的解读 (Szasz, et al., 2022, comment on *PNAS*)

- 对效应量的解读关系到科研工作者的预期，甚至会影响到整个领域后续研究的实验的设计和数据分析。
- **研究者需要认识到对研究“一直能得到持续且巨大的效应量”的期望是一种谬误。**



推荐阅读



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评述

迈向“发展群体神经科学”专辑



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开放式荟萃分析的规范化报告

刘宇¹, 陈树铨², 樊富珉¹, 邱新³, 范会勇⁴, 封春亮⁵, 郭双双¹, 甘怡群⁶, 李会杰^{7,8},
吕小康⁹, 任志洪¹⁰, 徐鹏飞¹¹, 袁博¹², 左西年^{13,14*}, 胡传鹏^{15*}

发表元分析的权威期刊推荐：

中文：心理学报，心理科学进展

英文：Psychological Bulletin

Educational Psychological Review

JPSP

https://bookdown.org/MathiasHarrer/Doing_Meta_Analysis_in_R/



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正确应用元分析：既不能夸大元分析的作用，
也不能否定元分析的用途。



Thanks !

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Helen Liu

HelenLiu0609

Social decision making, meta-research,
computational modelling | Diving, coffee,
cat and garden lover.

Free文章链接:

https://www.researchgate.net/publication/374007476_The_effectiveness_of_self_A_meta-analysis_of_using_self-referential_encoding_techniques_in_education

数据代码:

https://github.com/HelenLiu0609/SPE_edu_meta



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